

CONSTRUCTION OS

PRD_5

Role Context and Shared Module Navigation

High-Priority Full-Stack Bug Specification

Document	PRD_5
Status	Developer Implementation Specification
Priority	High
Compatibility	Additive to PRD_1, PRD_2, PRD_3 and PRD_4
Scope	All role dashboards and all shared modules

Implementation principle: The authenticated user's active role determines the dashboard shell. Permissions determine visible data and available actions. The opened module determines page content. A module must never switch the user to another role automatically.

1. Issue title

Role dashboard context is lost when users open shared modules such as Clients.

2. Problem summary

Construction OS contains modules that are accessible to multiple role types, including Sales, Finance, Legal, Architecture, Project Management, Executive users and other authorized roles.

Currently, when a user opens a shared module, the system redirects the user into the dashboard layout associated with another role.

Example: a Finance user logs in, opens Clients from the Finance sidebar, and the Clients page loads inside the Sales dashboard. The Finance user loses the Finance sidebar, Finance navigation context and Finance dashboard structure.

This problem occurs across all role types and must be corrected globally rather than patched only for the Clients page.

3. Current incorrect behaviour

Finance Dashboard

- > Finance user clicks Clients
- > Clients page opens
- > Dashboard shell changes to Sales

4. Required behaviour

Shared modules must open inside the currently authenticated user's active dashboard context.

A Finance user must remain inside the Finance dashboard shell. A Sales user opening the same Clients module must remain inside the Sales dashboard shell. An Architect must remain inside the Architecture dashboard shell.

The page content may be shared, but the surrounding dashboard context must remain role-specific.

Finance Dashboard

- > Finance user clicks Clients
- > Clients page opens inside Finance Dashboard
- > Finance sidebar and Finance navigation remain active

5. Core development rule

A module must not determine the dashboard layout. The authenticated user's active role or workspace must determine the dashboard layout.

The Clients route should render the reusable Clients module inside the dashboard shell resolved from the active session context.

Incorrect logic:

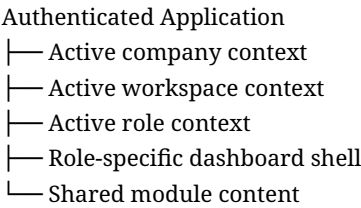
```
if (route === "/clients") {  
  return SalesDashboardLayout;  
}
```

Required logic:

```
return getDashboardLayout(  
  authenticatedUser.activeRole,  
  authenticatedUser.activeWorkspace  
);
```

6. Required architecture

Layer	Responsibility
Dashboard shell	Persistent role-specific sidebar, header, role identity, shortcuts, notifications, company/workspace context and permission-aware menu.
Shared module content	Reusable pages such as Clients, Projects, Tasks, Documents, Contractors and Meetings.
Resolution rule	Role/workspace resolves the shell; route resolves the content; permissions resolve data and actions.



7. Example dashboard behaviour

Role	Example sidebar	When Clients is opened
Finance	Finance Overview, Budgets, Expenses, Invoices, Payments, Clients, Contractors, Reports	Finance shell remains active; Clients is highlighted.
Sales	Sales Overview, Leads, Clients, Units, Reservations, Sales Reports	Sales shell remains active; Clients is highlighted.
Architecture	Architecture Overview, Drawings, Design Tasks, Clients, Projects, Approvals	Architecture shell remains active; Clients is highlighted.

8. Role-specific content inside shared modules

The Clients module may display different data, tabs and actions depending on the user's permissions. Dashboard context and content permissions are separate concerns.

The same Client Profile component may be reused while field visibility, tabs and actions are resolved from permissions.

Role	Possible client information/actions
Sales	Contact information, sales status, interested units, reservations, preferences, comments and requests.
Finance	Invoices, payments, outstanding balance, payment schedule and approved financial documents.
Legal	Contracts, amendments, notices, signatures and compliance status.
Architecture	Design requests, approved layouts, floor-plan requests, design comments and variations.

9. Active role context

The frontend must maintain an explicit active-role context. The context must survive internal navigation, refresh, browser history, notifications, global search, related-record links and direct URL access.

```
interface ActiveUserContext {
  userId: string;
  companyId: string;
  workspaceId: string;
  activeRoleId: string;
  activeRoleCode: string;
  departmentId?: string;
  permissions: string[];
}
```

10. Route handling

Shared routes should not be owned by one role. The team may use role-neutral routes with a session-resolved shell, or role-prefixed routes that only represent dashboard context. The underlying shared module must remain reusable.

The routing strategy must be applied consistently across the application.

Preferred role-neutral routes:

```
/app/clients
/app/clients/:clientId
/app/projects
/app/projects/:projectId
/app/tasks/:taskId
```

Acceptable role-context routes:

```
/app/finance/clients
/app/sales/clients
/app/architecture/clients
```

11. Direct URL behaviour

When a Finance user opens a direct Client Profile URL, the application must authenticate the user, resolve company and active role, validate access, load the Finance shell and render the profile inside it.

The system must not infer that the user is a Sales user because the record belongs to the Clients module.

12. Multiple-role users

Employees may hold more than one role. Shared modules must use the currently selected active role. Opening a module must never change the active role automatically.

13. Sidebar requirements

The original role-specific sidebar must remain visible, the shared module must be highlighted, unauthorized items must remain hidden, and the expanded or collapsed sidebar state should remain unchanged.

```

Finance
├─ Overview
├─ Budgets
├─ Expenses
├─ Invoices
├─ Payments
├─ Clients  <- Active
├─ Contractors
└─ Reports
  
```

14. Breadcrumb requirements

Breadcrumbs must preserve dashboard context. The same client may display as Finance / Clients / Client Name for Finance and Sales / Clients / Client Name for Sales.

15. Permission requirements

Preserving the user's dashboard shell must not grant additional access. Backend authorization must continue to validate company, workspace, role, department, record and field-level permissions.

Frontend route protection is not sufficient. APIs must return only data authorized for the authenticated user.

16. Possible technical cause

The exact cause is unknown. The team must investigate layout nesting, direct Sales layout imports, default route middleware, missing role state, Sales-prefixed links, hard-coded sidebar configuration, page refresh fallbacks and links that omit role or workspace context.

These are investigation points and not confirmed causes.

17. Required fix scope

The fix must apply to all navigation paths: sidebar links, dashboard cards, notifications, global search, recent items, favorites, related records, timelines, task links, direct URLs, page refresh and browser history.

18. Test matrix

The team must test every shared module against every role that is authorized to access it. The same matrix must be repeated for Projects, Tasks, Documents, Contractors, Meetings, Reports and all other shared modules.

Role	Can open Clients	Expected shell
Sales	Yes	Sales
Finance	Yes	Finance
Legal	Yes	Legal
Architecture	Yes	Architecture
Project Manager	Yes	Project Management
CEO	Yes	Executive
Company Admin	Yes	Admin

19. Acceptance criteria

1. A Finance user can open Clients without being redirected to Sales.
2. The Finance sidebar remains visible on the Clients list and individual Client Profile.
3. Sales, Architecture, Legal and all other roles remain inside their own shells.
4. The selected shared module is highlighted in the correct sidebar.
5. Browser refresh and browser back/forward preserve active role context.
6. Direct URLs, notifications and global-search results open inside the active role shell.
7. Users with multiple roles remain in the currently selected role.
8. Shared pages do not import, force or default to a specific role layout.
9. Backend permissions remain enforced for every API response.
10. No duplicated Clients module is required for each role.
11. Automated tests cover role context for all shared routes and authorized roles.

20. Expected final result

Construction OS must behave as one operating system with role-specific workspaces. Users may access the same shared company records, but they must remain inside their active role dashboard unless they explicitly switch roles.

Authenticated role determines dashboard shell.

Permissions determine available data and actions.

The opened module determines page content.

The module must never change the user's role automatically.

Classification: High-priority navigation and role-context defect affecting every dashboard, shared module and user role.